

Seacraft

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海舟智能

Cloud-Edge-Terminal Collaboration in Higher Education Scenarios

Cloud-Edge-Terminal Architecture

Integrated Video Middle Platform

"4+X" Smart Classroom

Digital Teaching Resource Platform



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Seacraft Technology

Suzhou Seacraft Technology Co., Ltd. (hereinafter referred to as "Seacraft Technology") is an independent education industry company incubated by Suzhou Keda Technology Co., Ltd. (Stock Code: 603660), formerly known as the Education Division of Suzhou Keda. The company name "Seacraft" implies "Endless learning is like sailing a boat through the sea," symbolizing the spirit of continuous exploration and advancement in the ocean of digital education. Seacraft Technology is a leading provider of digitization and smart transformation solutions for higher education institutions in China, dedicated to empowering the high-quality development of higher education. The company has been deeply involved in the fields of audio-visual capture, processing, and AI analysis for many years, forming a comprehensive product and solution matrix that integrates "cloud, edge, and terminal" collaboration.

Teaching: Providing the "4+X" smart classroom solution, building multifunctional smart classrooms that are "clear to see and hear."

Learning: Offering solutions such as a live recording platform and digital teaching resources to aid in the digital transformation of teaching models.

Management: Creating a new integrated audio-visual resource management system through a converged video communication platform and AI digital class monitoring.

Evaluation: Building a visual teaching quality assessment system with AI analysis at its core, assisting in the reform and innovation of educational evaluation.

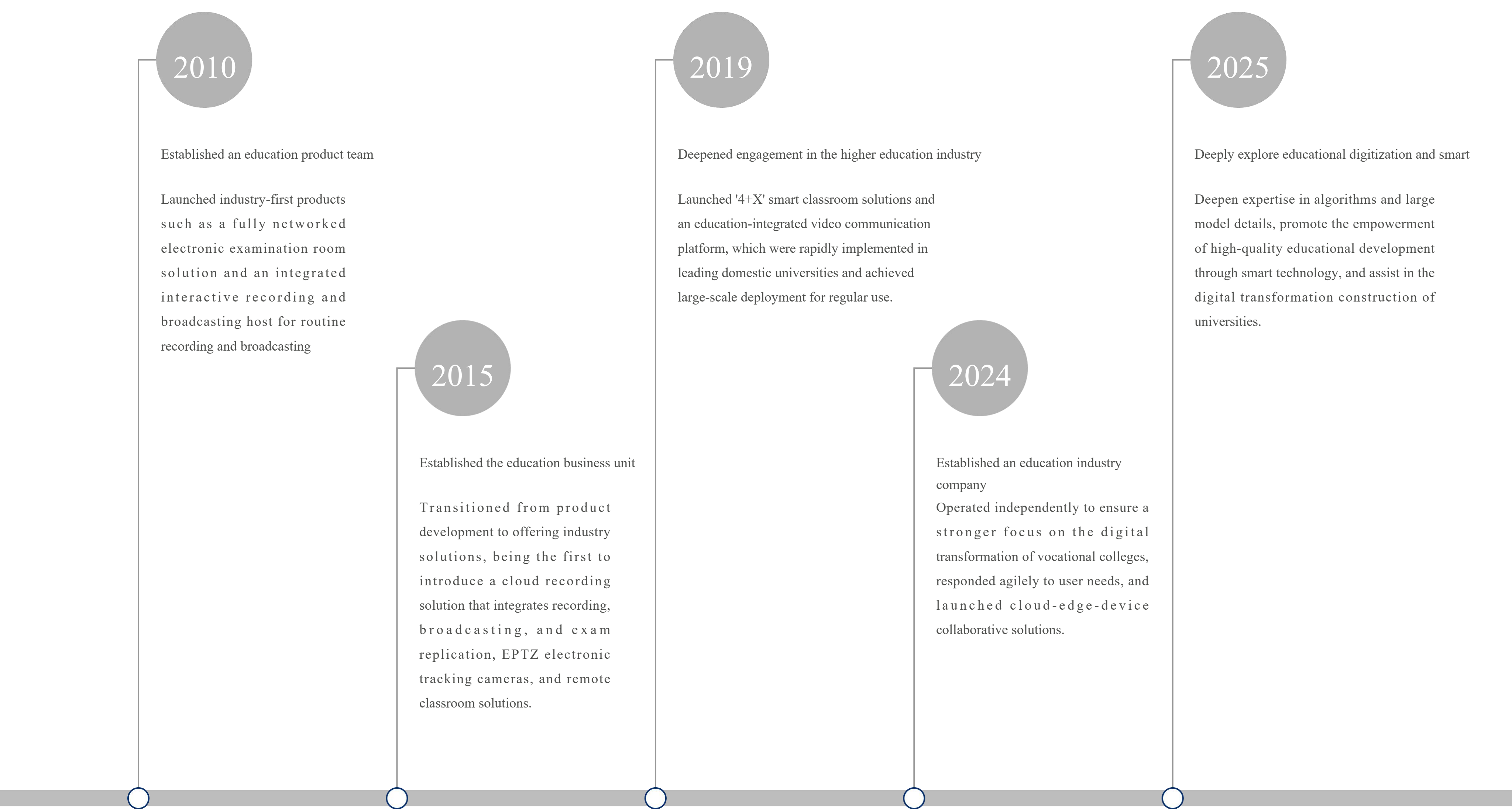
Examination: Deploying an AI-empowered exam monitoring system to construct a multimodal abnormal behavior warning system, ensuring the fairness of exams.

In the future, Seacraft Technology will continue to focus on the core needs of digital transformation in higher education institutions, driven by technological innovation and guided by application scenarios. The company aims to provide strong technical and service support for the digital construction of higher education, promoting the creation of a digital campus that prioritizes moral education as its foundation, data governance as its cornerstone, and human-computer collaboration as its distinctive feature, making every class more exciting!

Vision: Make Every Class More Exciting

Mission: Empower Digital Education with Technology and Service

Values: Continuous Innovation, Creating Value for Customers and Employees



Cloud-Edge-Terminal Architecture

Through the seamless collaboration of cloud services, cloud platforms, edge computing, and terminal acquisition devices, an efficient, flexible, and smart distributed collaborative architecture is constructed. The cloud serves as the brain, providing powerful computing capabilities, data storage, and business applications. The edge acts as a hub, responsible for partial real-time data processing and local decision-making, reducing transmission latency and enhancing local response speed. Terminal devices function as sensing and execution nodes, collecting data and executing commands to achieve instant interaction. Through unified protocols and data interfaces, the three work in harmony to realize resource sharing, task balancing, and smart applications, meeting the digital transformation needs of universities and driving the deep integration and innovative development of university operations with cutting-edge technologies such as cloud computing, AI artificial smart, and big data.



Integrated Video Middle Platform

The integrated video middle platform, based on the concept of microservices and adopting a clustered deployment architecture, forms a private cloud video middle platform solution for campuses using Kubernetes and Docker. The platform can be deployed on the school's virtual machine clusters or physical machines, supporting high-availability deployment and elastic scaling. It allows flexible scheduling of the capabilities of various microservices within the cluster, empowering upper-layer applications on-demand and providing functional applications such as system management, integrated media services, AI algorithm engines, space management, and open platforms.

Access

Integrate existing systems
Fully utilize existing resources

Integration

Integrate media resources
to ensure consistency

Application

Merge business applications
to maximize value

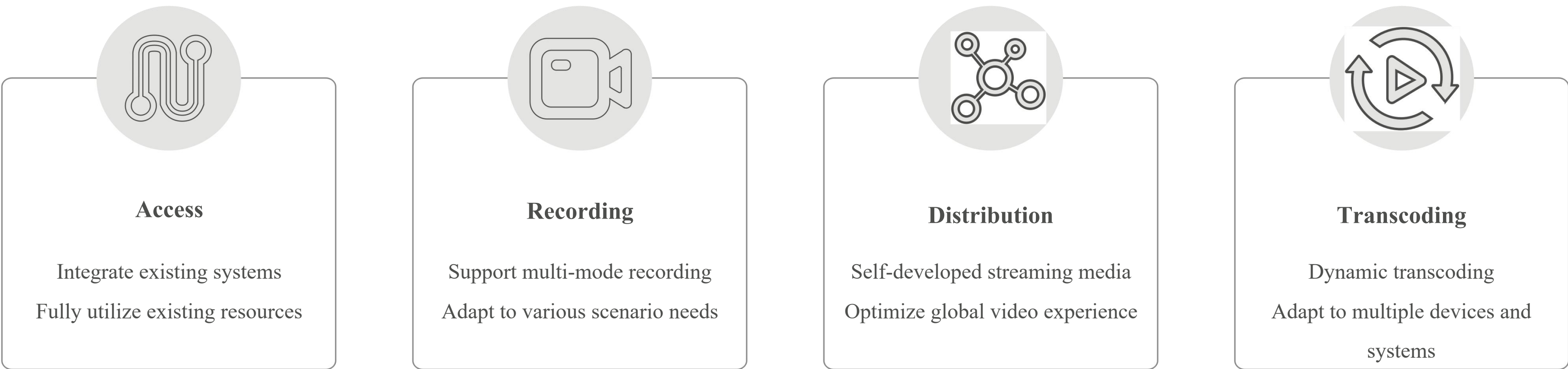
Open

Open architecture
Meet personalized needs



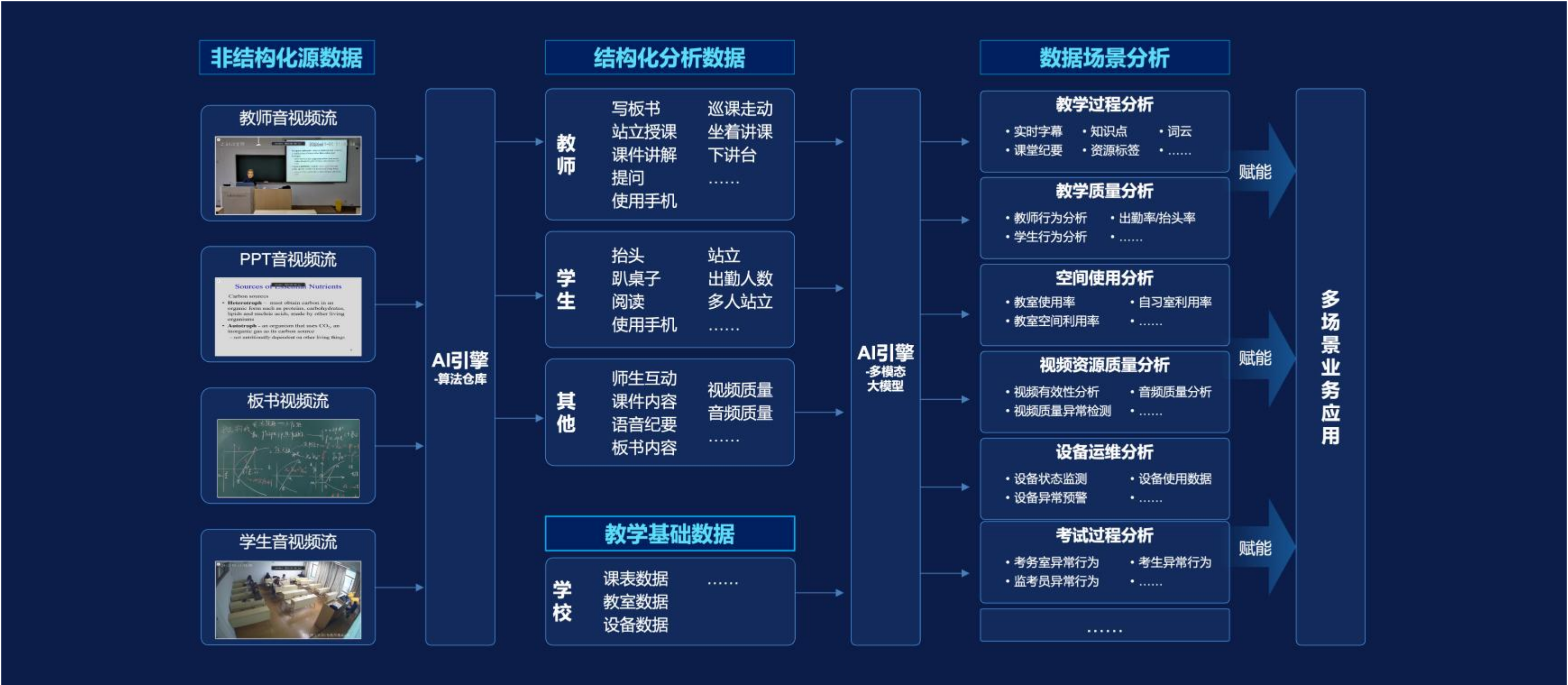
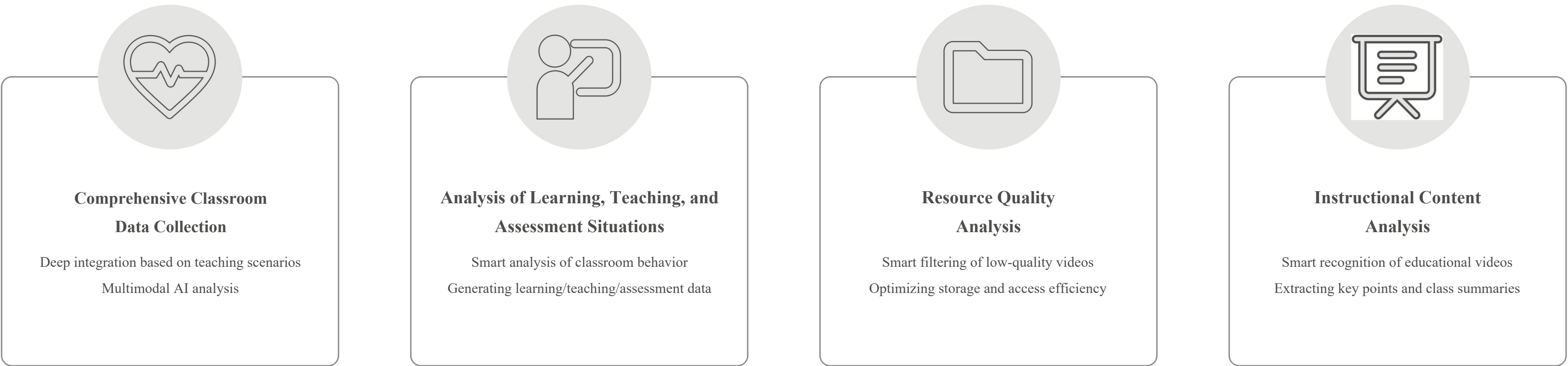
Converged Media Services

Converged media services are composed of microservices such as media access, media processing, and media scheduling, providing a comprehensive media management solution. It covers functions like device access, video storage, stream distribution, and transcoding optimization, supports multiple video protocols and storage modes, achieves efficient media resource integration and flexible scheduling, and meets the demands of high-concurrency browsing, multi-scenario recording, and cross-platform high-quality video experiences.



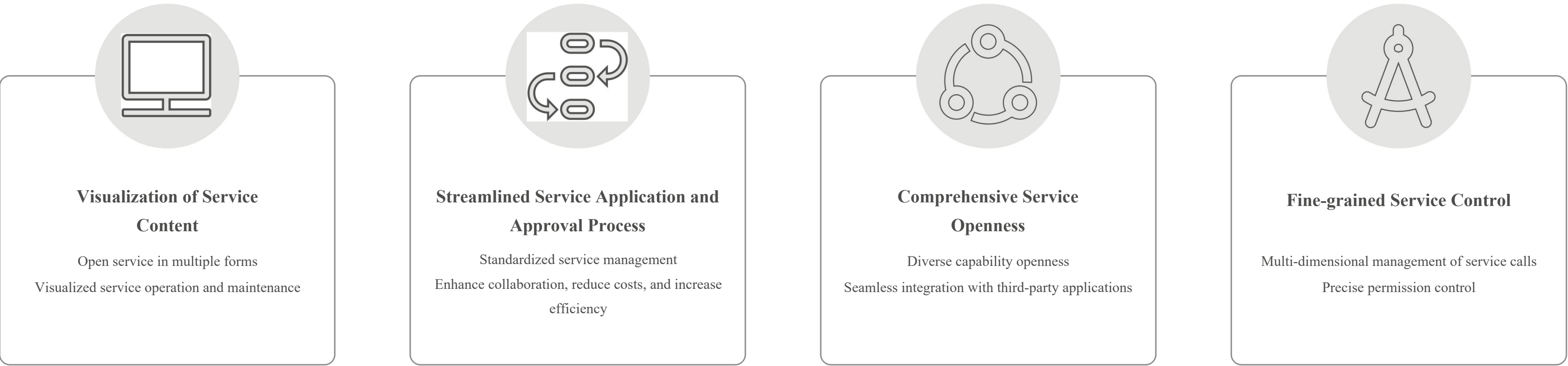
AI Algorithm Engine

Provide a multimodal AI analysis algorithm engine based on teaching scenarios, offering capabilities such as algorithm library calls, business model library calls, data processing, and model parameter adjustments, supporting schools' digital construction with computational power and algorithms. Built-in multiple algorithm repositories enable comprehensive AI smart analysis of video resources, providing robust data support for upper-layer applications.



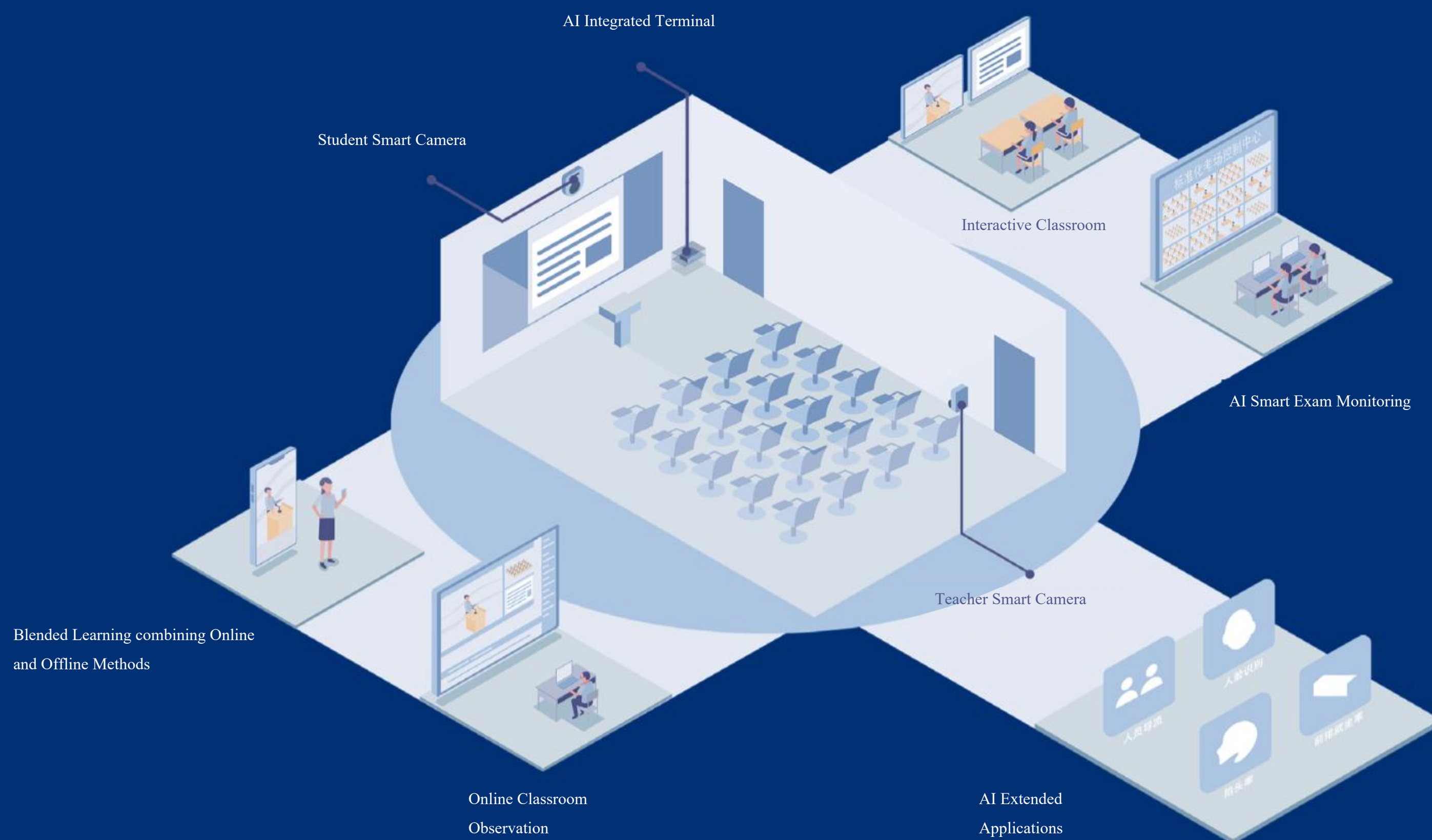
Open Platform

Through standardized open service APIs and WEB SDK components, as well as mature industry and technical solutions, unified video conferencing capabilities are opened to third-party business systems of the school in a standardized manner, improving collaboration efficiency, making service content visual, service application and approval processes streamlined, service openness comprehensive, and service control fine-grained.



'4+X' Smart Classroom

The construction of smart classrooms should adhere to the following principles: overall planning, phased construction, prioritizing infrastructure, and progressive application services. Seacraft Technology's '4+X' smart classroom solution requires only one set of equipment to meet all audio-visual application needs of a smart classroom, enabling four fundamental applications: hybrid teaching (online and offline), online classroom observation, interactive classrooms, and smart exam monitoring. Additionally, by leveraging the converged video communication platform to reuse front-end devices, it is possible to extend applications such as digital teaching resources, digital teaching evaluation, and digital management support. Through the cloud-edge-device collaborative architecture, comprehensive classroom element collection can be achieved, enabling 'X' extended applications such as AI voice transcription and translation, AI student/teacher/exam performance analysis, media asset management, classroom navigation, smart operation and maintenance, and digital twin.



Recording Terminal

Adhering to the philosophy of making it 'convenient and effortless' for teachers to use, and driven by the demands of regular recording scenarios, we continuously iterate and upgrade our products. This is achieved by integrating our company's core technological strengths in 'real-time interaction,' 'large-scale concurrency,' and 'simplicity and stability.' From product function design to software algorithm updates and hardware component selection, we provide users with a variety of easy-to-use recording terminals to choose from.



AI Convergence Terminal New Product

- AES67 digital audio, mixing and ducking, smart sound reinforcement
- Support AoIP broadcasting, seamless integration with broadcasting systems
- Built-in multi-protocol wireless screen sharing for flexible content sharing
- Support edge AI applications, enabling cloud-edge-device collaborative AI applications



High-Definition Interactive Lecture Recording Host

- Interactive recording and broadcasting integration, full aluminum body static heat dissipation
- 5A algorithm, meeting-level audio and video interaction effects
- Dual-channel 4K courseware capture, professional audio and video codec technology
- Dual UVC interface for hybrid teaching



Lightweight Lecture Recording Host

- Lightweight design, embedded hardware architecture
- Support UVC interface for hybrid teaching
- Support 4K courseware capture, automatic video composition and broadcasting
- Compatible with multiple protocols such as Onvif/RTSP/RTMP



Online Course Terminal

- Compact design, easy installation
- Support 4K courseware capture, supports audio noise reductionSupports
- UVC interface for hybrid teaching
- Support image synthesis, optional multiple screen layouts

Camera

Adhering to the philosophy of ensuring students can 'see clearly and comfortably' during online learning, and considering the varying brightness levels of projectors and blackboards in university classroom settings, we have continuously iterated and upgraded our products in terms of aesthetic design, software algorithm updates, and lens selection to meet the multifaceted requirements of smart classroom construction. This includes providing users with a variety of camera options featuring different functionalities.



Teacher Smart Camera

- 18x optical zoom, suitable for various medium and large classrooms
- Multi-stream output: teacher close-up, whiteboard close-up, podium panorama
- Gamma adaptive, balancing high-light projection with whiteboard and facial clarity
- Whiteboard enhancement and whiteboard extraction make learning more convenient for students



Teaching Smart Camera

- Smart teacher and student tracking, clear whiteboard extraction
- Four-channel stream output, seamless multi-view switching
- Ultra-wide dynamic correction, ensuring stable and distortion-free visuals
- Two-way audio noise reduction, facilitating convenient local storage



Teacher Smart Camera

- Smart dual-lens camera, full coverage of teaching scenarios.
- High-definition dynamic imaging, outstanding in detail clarity.
- Smart tracking and switching, smoother classroom interaction.
- Ultra-wide-angle view, capturing every exciting moment.



Student Dual-lens Camera

- Dual-lens design, covering both panoramic and close-up shots
- Dual network port design, ensuring the security of exam data
- Student face capture, standing tracking, precise learning analysis
- Wide-angle lens, adaptable to various classroom types like seminar and U-shaped



Pan-tilt Camera

- Multiple zoom lenses available to meet diverse needs
- Flexible gimbal rotation for capturing panoramic classroom dynamics
- Cloud-based flexible control for free perspective switching
- Smart detection function to comprehensively safeguard classroom safety



Panoramic Camera

- 50-meter infrared + Smart IR for smart day and night shooting
- 120dB wide dynamic range for shooting in backlit environments
- Built-in Mic and Speaker supporting two-way intercom
- Support 256G TF card for local backup storage

Digital Audio

Adhering to the philosophy of 'Clarity, smart, and Seamlessness,' we address the audio needs of various types of classrooms by comprehensively considering core challenges such as environmental noise suppression, near/far-field distance, and speech clarity. By integrating lightweight design, smart algorithms, and convenient deployment requirements, we continuously innovate across multiple dimensions including hardware performance optimization, audio processing technology upgrades, and breakthroughs in smart noise reduction algorithms. This enables us to provide users with high-fidelity sound quality, precise voice capture, and flexible audio processing and pickup solutions.



Digital Audio Processor

- Support multi-channel smart mixing, audio ducking, and smart noise reduction
- Full-duplex AEC algorithm for precise echo cancellation
- Adaptive Feedback Cancellation (AFC) algorithm meets long-distance sound reinforcement needs
- Built-in 2*125W amplifier, high integration, one-click debugging



Smart Mixer

- Compact and lightweight, empowering professional sound quality
- Built-in embedded microphone for clear sound pickup
- Smart ducking for smooth, interference-free audio effects
- Precision mixing for clearer and more outstanding sound



Digital Array Mic

- 360° omnidirectional + 3 x 120° super-cardioid
- Smart 3A algorithm audio processing technology
- 9-meter pickup range, 3 levels of daisy-chain cascading
- Voice enhancement for 'comfortable listening'



Network Array Mic

- 8-array beamforming algorithm for precise voice capture
- Smart noise reduction + voice enhancement for pure and natural sound
- Support POE power supply, single network cable connectivity
- AES67 digital audio for high-fidelity transmission

Servers & Storage

Adhering to the philosophy of 'efficiency, reliability, and security,' we address the diverse needs of university operations by considering challenges in high-performance computing, massive data storage, and business continuity across multiple scenarios. Combining requirements for modular design, smart management, and energy efficiency, we continuously innovate from multiple dimensions, including hardware architecture optimization, software algorithm upgrades, and storage technology breakthroughs. This provides users with flexible, scalable, and secure server and storage solutions that meet the deployment needs of businesses of various scales.



High-Density Blade Servers

- High-density design supporting multi-node parallel computing
- Fully modular hot-swappable components for easier maintenance
- Support virtualization deployment and is compatible with mainstream operating systems
- High-efficiency power supply for energy-saving and environmental protection



Disk Storage Array

- Cable-free modular design, stable and easy to maintain.
- 3U high-density rack for efficient space utilization.
- KRAT technology ensures data security.
- Streaming media scheduling makes resource management smarter.



Cloud Storage

- Dual 10GbE ports for efficient read/write and load balancing.
- Erasure code redundancy protection ensures data integrity even with multiple disk failures.
- Modular cable-free design enables convenient and reliable maintenance.
- Flexible storage for massive data with scalable performance and capacity.



General Purpose Large Model All-in-One Machine

- Integrated hardware and software, ready to use out of the box
- Universal architecture design, exceptional cost-effectiveness
- High-performance GPU configuration, supports elastic computing power expansion
- Pre-installed multimodal algorithms, meets AI needs across various scenarios



Domestic Large Model All-in-One Machine

- Built-in DeepSeek R1/V3, ready to use out of the box
- Full-stack domestic support, meets national innovation requirements
- Flexible computing power card configuration, supports elastic computing power expansion
- Pre-installed multimodal algorithms, meets AI needs across various scenarios



Domestic Media Service All-in-One Machine

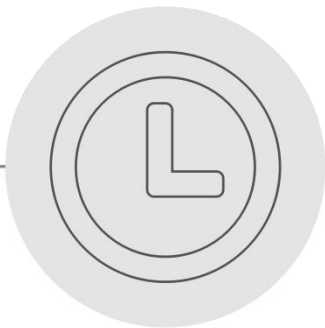
- Powered by domestic chips, safeguarding the value of every piece of data
- Support multi-protocol access, handles large concurrent resource recording
- Direct MP4 format storage, seamless course recording
- High concurrent media access, significantly enhances video stream efficiency

Digital Teaching Evaluation

The digital teaching evaluation system constructs a comprehensive, traceable, and closed-loop teaching quality assurance system for schools. It enables precise tracking of various teaching data such as classroom instruction, course quality, and teacher development, and supports smart configuration and adjustment of evaluation indicators. The system includes a multi-terminal evaluation system compatible with PCs, tablets, large screens, and mobile devices, offering core functionalities such as online class observation and evaluation, multi-window class inspection, and analysis of student and teaching conditions. It is widely applied in teaching scenarios such as leadership class observation, teaching supervision, peer review, audit evaluation, and professional development learning.



Digital Classroom Observation and Evaluation



Online Real-time Classroom Observation and Evaluation

If the course is being live-streamed, the observing experts can watch the course in real-time online and complete the evaluation.



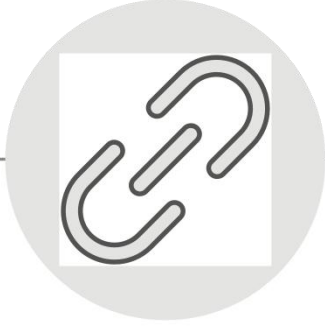
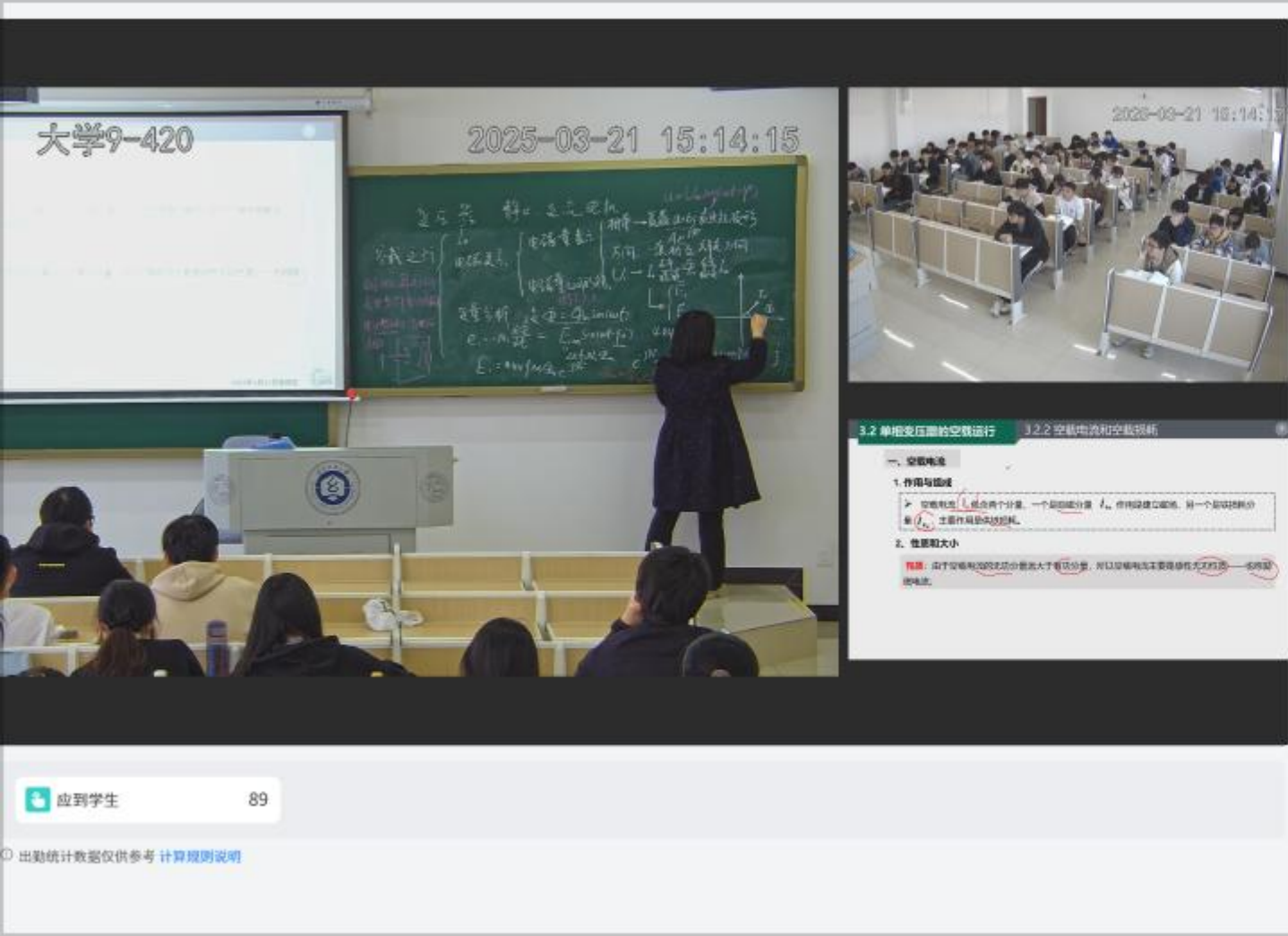
Online Post-class Observation and Evaluation

After the course ends, listening experts can watch the course replay video online and complete the course evaluation.



Offline Course Listening and Evaluation

Supports offline course listening, online filling of supervision evaluation forms, uploading evaluation data to the system, and achieving unified online and offline management.



Automatic Matching of Different Plans

According to expert permissions and plan permissions, automatically supports matching of university-level plans, college-level plans, and special evaluation plans.

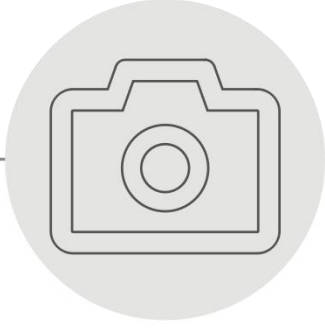
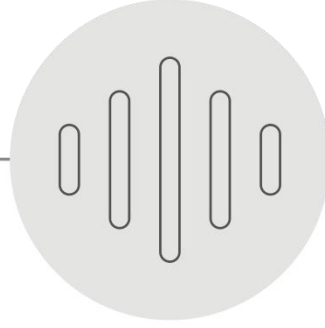


Photo Commentary

Supports capturing classroom video frames and taking screenshots with explanations, allows marking outstanding segments, and aggregates them to the Classroom Observation Center

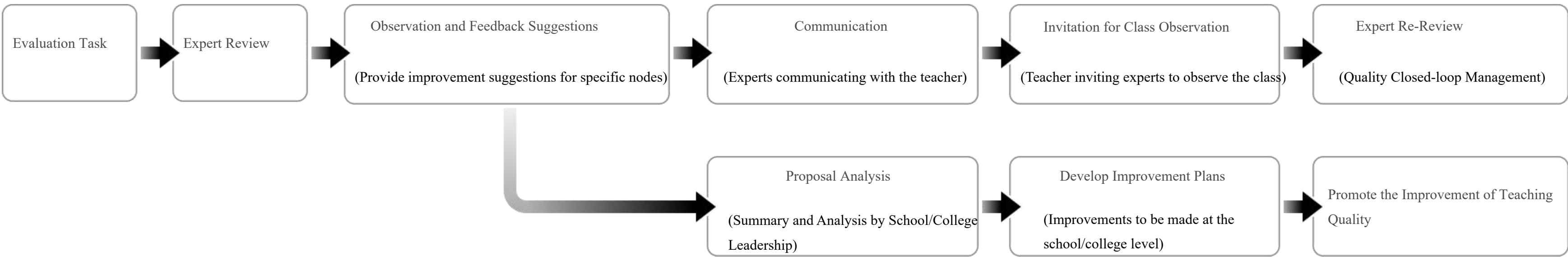


AI-Assisted Evaluation

AI-powered learning and teaching analysis to quickly understand classroom conditions across the school and improve observation efficiency

Expert Feedback Loop

During the evaluation process, expert teachers can send questions or suggestions from the course to the teacher being evaluated, the teaching department, or college-level leadership.

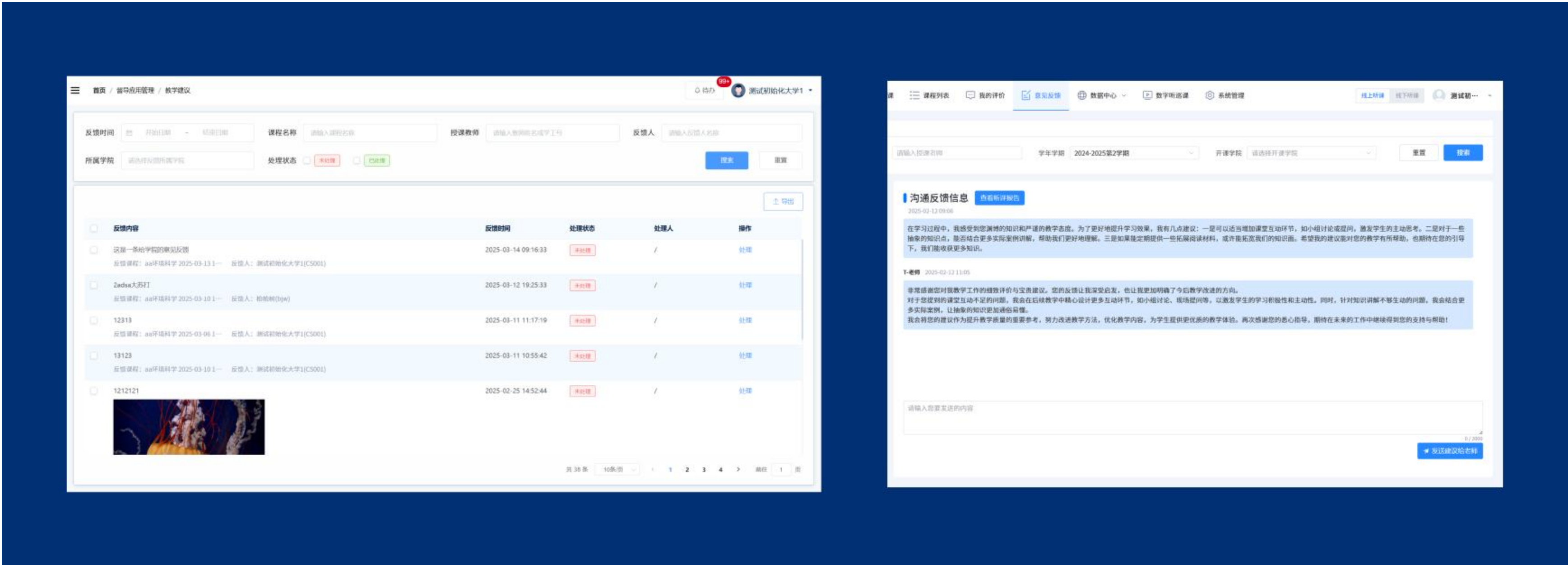


Experts and Teachers

Effective communication and interaction can be established between experts and teachers. Teachers who have been observed can invite experts for a second round of evaluation, creating a positive validation mechanism that helps improve classroom and course quality.

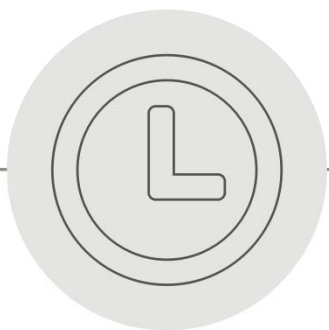
Experts and Institutions

School administration offices and department leaders can summarize and analyze the feedback submitted by experts, formulate improvement plans, and promote the enhancement of teacher competencies and course quality.



AI Classroom Analysis

Support multi-dimensional analysis based on learning data, viewing duration, teacher behavior, and student behavior, deeply interpreting classroom data. Provide precise teaching feedback and personalized learning suggestions for university users, assisting teachers in optimizing course design and improving teaching efficiency. Driven by accurate data, empower teaching quality evaluation and enhancement, safeguarding the cultivation of high-quality talent!



Classroom Analysis Report

Supports one-click generation of classroom analysis reportsGain insights into classroom data, teaching quality at a glance



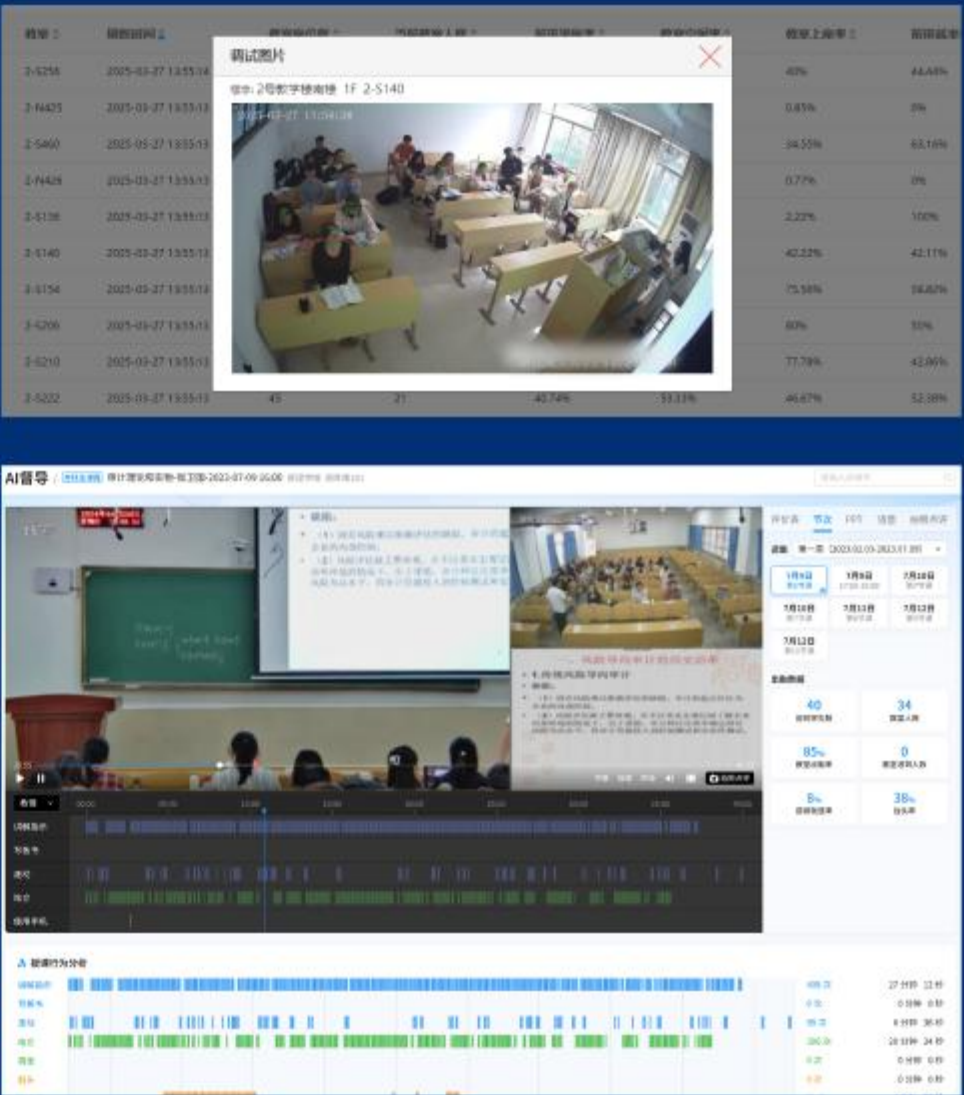
Teacher Behavior Analysis

Identify and detect various teaching behaviors of teachers, providing data for teaching management and teaching quality evaluation.



Student Behavior Analysis

Identify and detect various teaching behaviors of students, providing data for teaching management and teaching quality evaluation.



Digital Teaching Resource Platform

“ The 'Galaxy' Digital Teaching Resource Platform enables schools to establish a one-stop personalized video resource portal. It facilitates the unified management of various teaching video resources such as classroom recordings, top-quality courses, public lectures, keynote speeches, major events, news reports, and personal videos. The platform supports smart hierarchical access control, emphasizes intellectual property protection, and integrates multimodal large models and smart audio-video analysis technologies to achieve structured data extraction, user profiling, resource profiling, and personalized resource recommendations, thereby building a personalized and digital future learning center for schools.



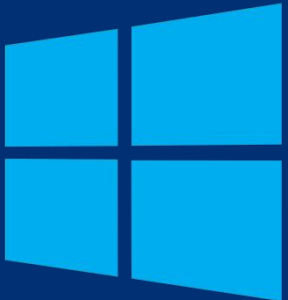
Triple-end Multi-system Compatibility

Android



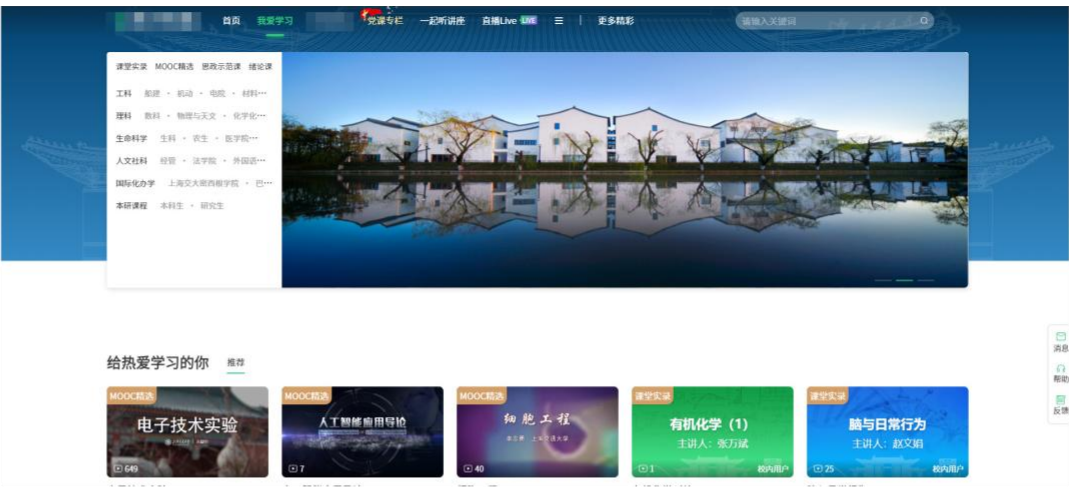
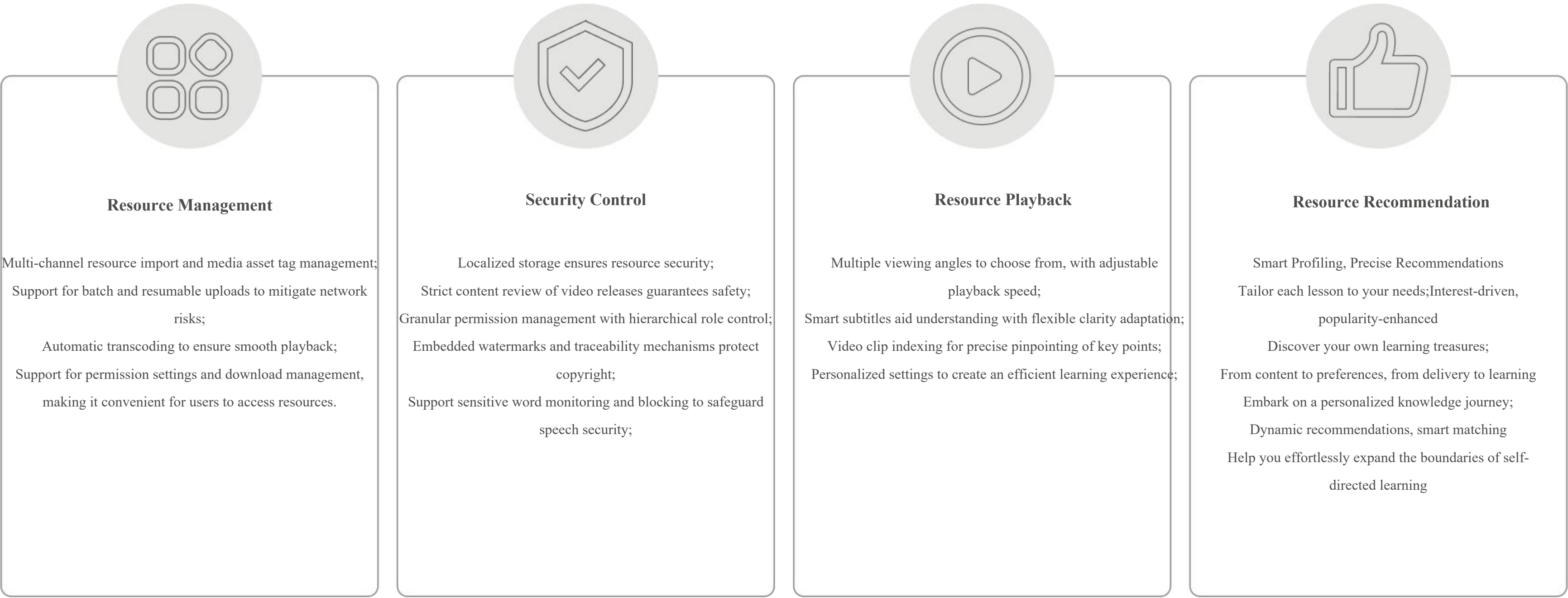
iOS

HarmonyOS



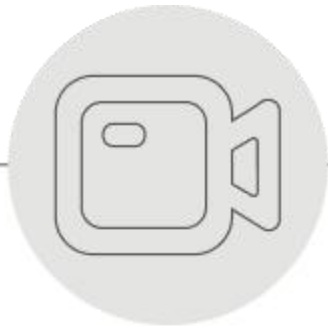
Video Portal

Aggregate various video resources from the school, allowing users to browse and access based on their permissions and needs. Customizable school-specific sections and themes are supported, with multi-dimensional filtering and browsing options. Various forms of interaction are provided during video playback.



Classroom Recording

The classroom recording platform focuses on the core business of routine classroom recording, primarily offering smart live and on-demand course functions for all students, as well as recording management and data dashboard features for teachers.



Routine Recording

Support self-service recording and automatic recording based on schedules

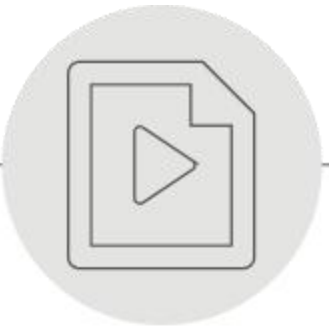
Host recording and cloud recording dual solutions

Remote device and business management

Easy operation and maintenance of multi-angle live streaming;

Support main and secondary screen directing, AI real-time subtitles, etc.;

Support smart summarization, OCR text extraction, and other functions to meet personalized learning needs.



Record Management

Support course attribute editing and permission settings,

Flexibly control course access scope;

Provide course video sharing functionality,

Support viewing record queries and courseware material management;

Support voice subtitle publishing;

Support classroom note annotation;



Data Dashboard

Provide real-time and cumulative view data,

Comprehensively grasp the popularity of courses;

Support statistics and ranking of play counts by college and subject,

Form multi-dimensional ranking analysis;

Support analysis of user access terminals, time periods, and trends,

Gain an in-depth understanding of user viewing habits;

Support direct detection of sensitive words in live and on-demand content,

Implement public sentiment control;

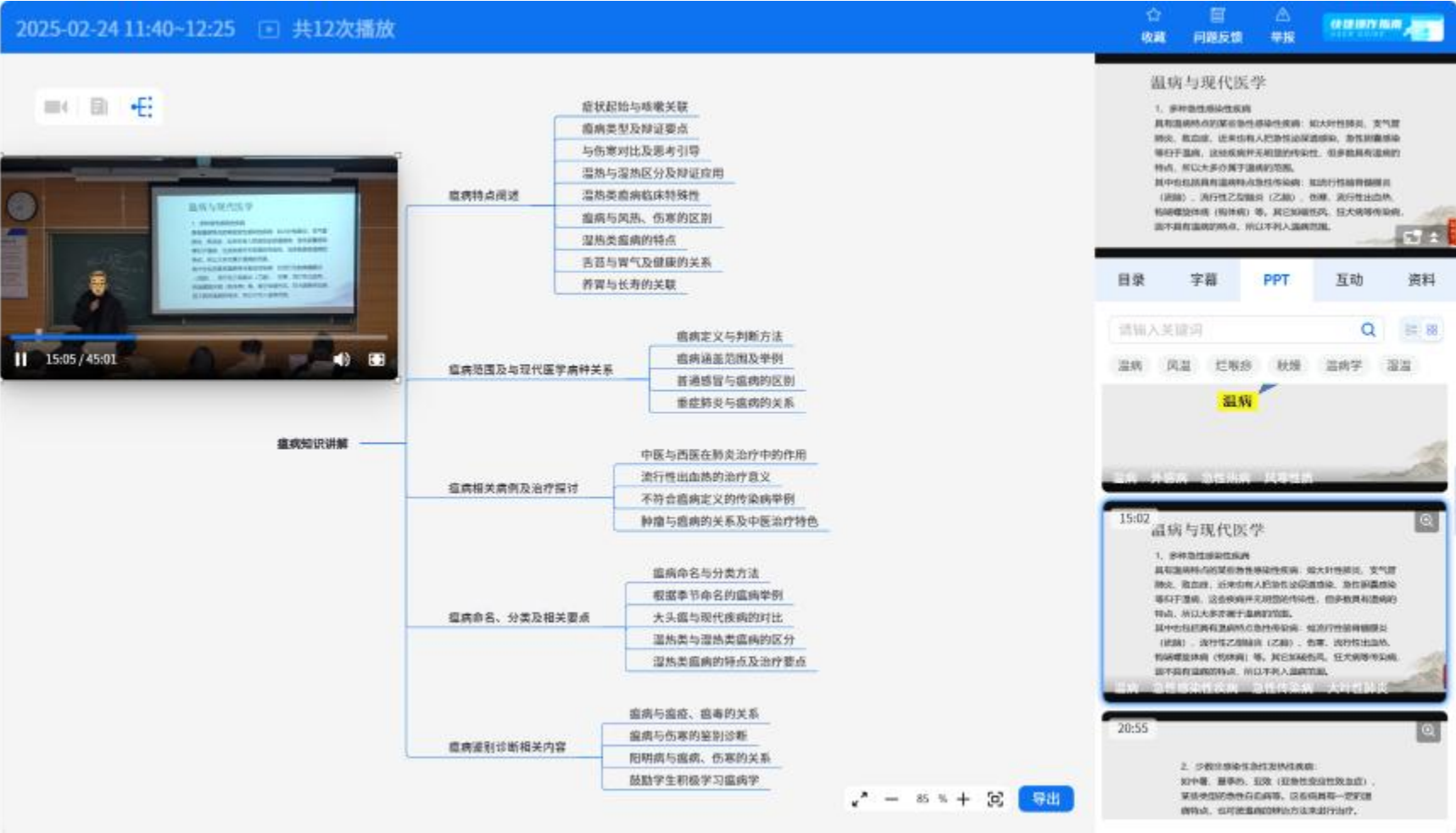


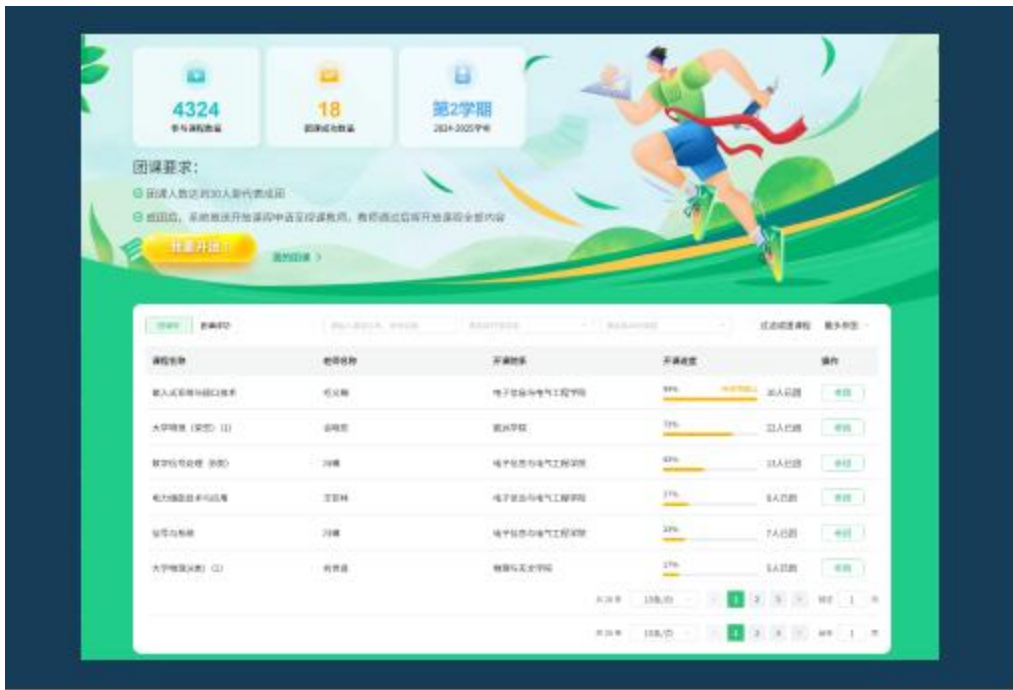
Resource Quality Analysis

Automatically identify video resources with unclear images, incorrect formats, or playback issues to ensure file quality;

Accurately determine whether videos contain effective class information and filter out meaningless video content;

Analyze resource utilization and filter out idle videos;





Preview Of Introductory Courses and Group Class Mechanisms

Select one recorded lecture from each course as an introductory session open to all faculty and students to reduce dropout rates. Students who did not secure a spot in popular courses can initiate or participate in group classes, and teachers can open courses based on student demand, fostering autonomy and helping the school understand student interests and teaching evaluations.



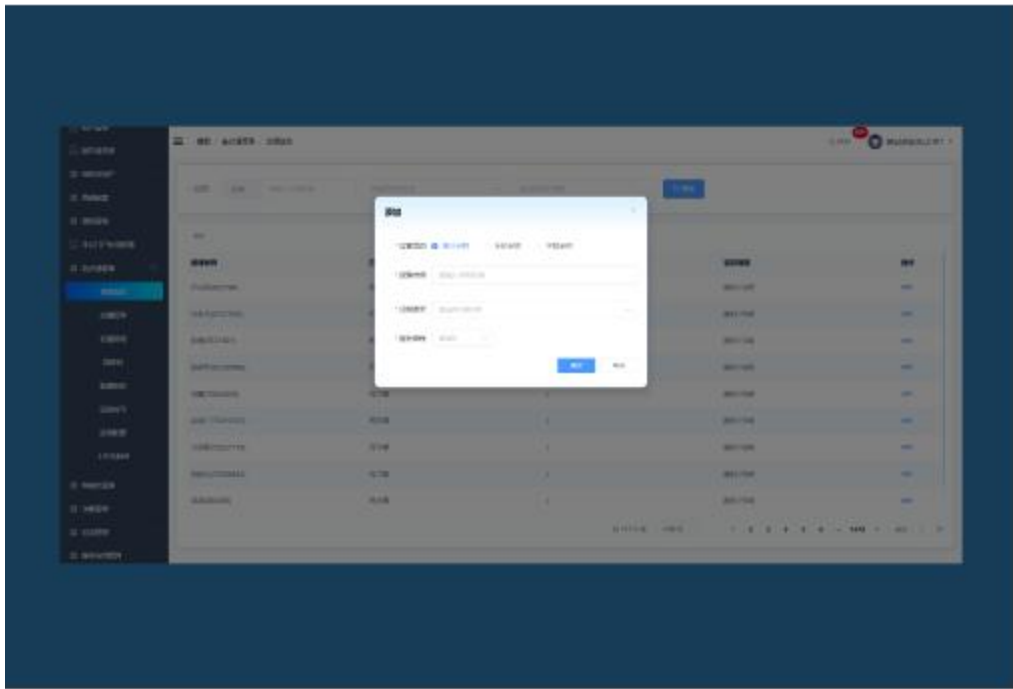
Event Livestream LIVE

Major on-campus event livestreams can be showcased through an independent page module, offering features like pre-event announcements, reservations, and notifications, and supporting third-party streaming and CDN acceleration. Post-livestream, on-demand resources are automatically generated, available for re-editing or immediate viewing, meeting the needs of open lectures, seminars, and more.



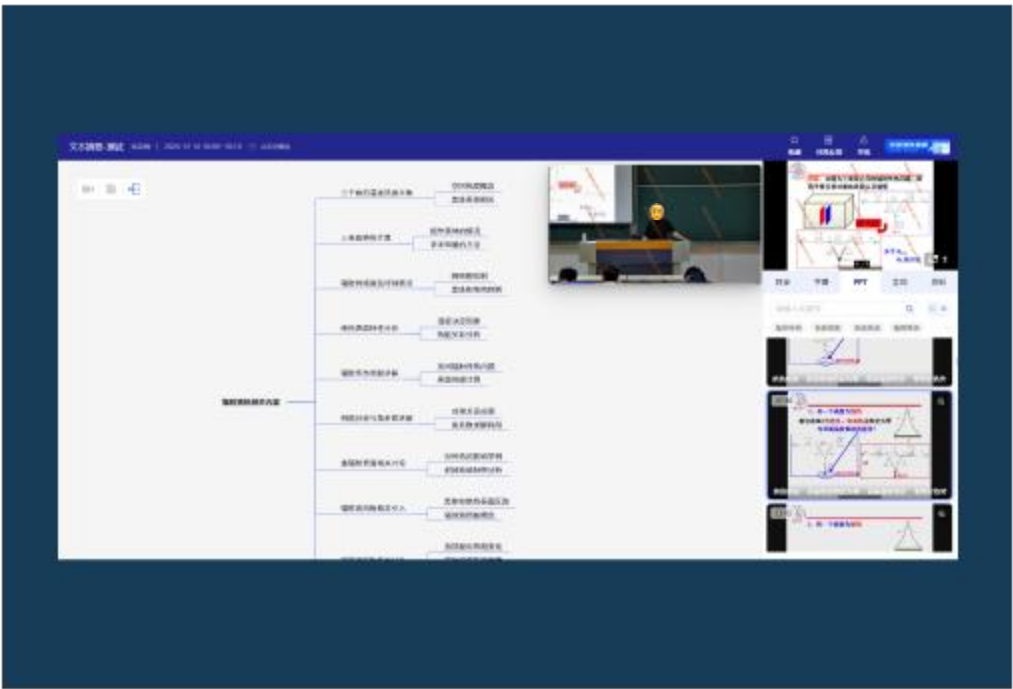
Smart Playback Module

Supports note-taking, bookmarking, and jump-to-point features to enhance learning efficiency; interactive comments and smart management add fun; no-plugin playback, speed adjustment, and watermarking ensure a smooth experience; integrated mind maps and floating progress bars make key points clearly visible. Additionally, it supports PPT segmentation and OCR recognition, offering flexible and easy-to-use resources for precise and efficient learning!



Differentiated Retention Mechanism

Provide a differentiated retention mechanism for video resources. For regular courses, a periodic coverage can be set, while for classic videos with high popularity and quality, or important videos reused for teaching competitions, teaching evaluations, etc., they can be set for long-term retention. Through this differentiated retention mechanism, storage pressure is reduced.



Knowledge Point Index

Generate course knowledge points and mind maps through multimodal AI algorithms and models, effectively summarizing and organizing knowledge from videos. This enhances the review efficiency for both teachers and students, establishing a solid data foundation for building knowledge graphs!

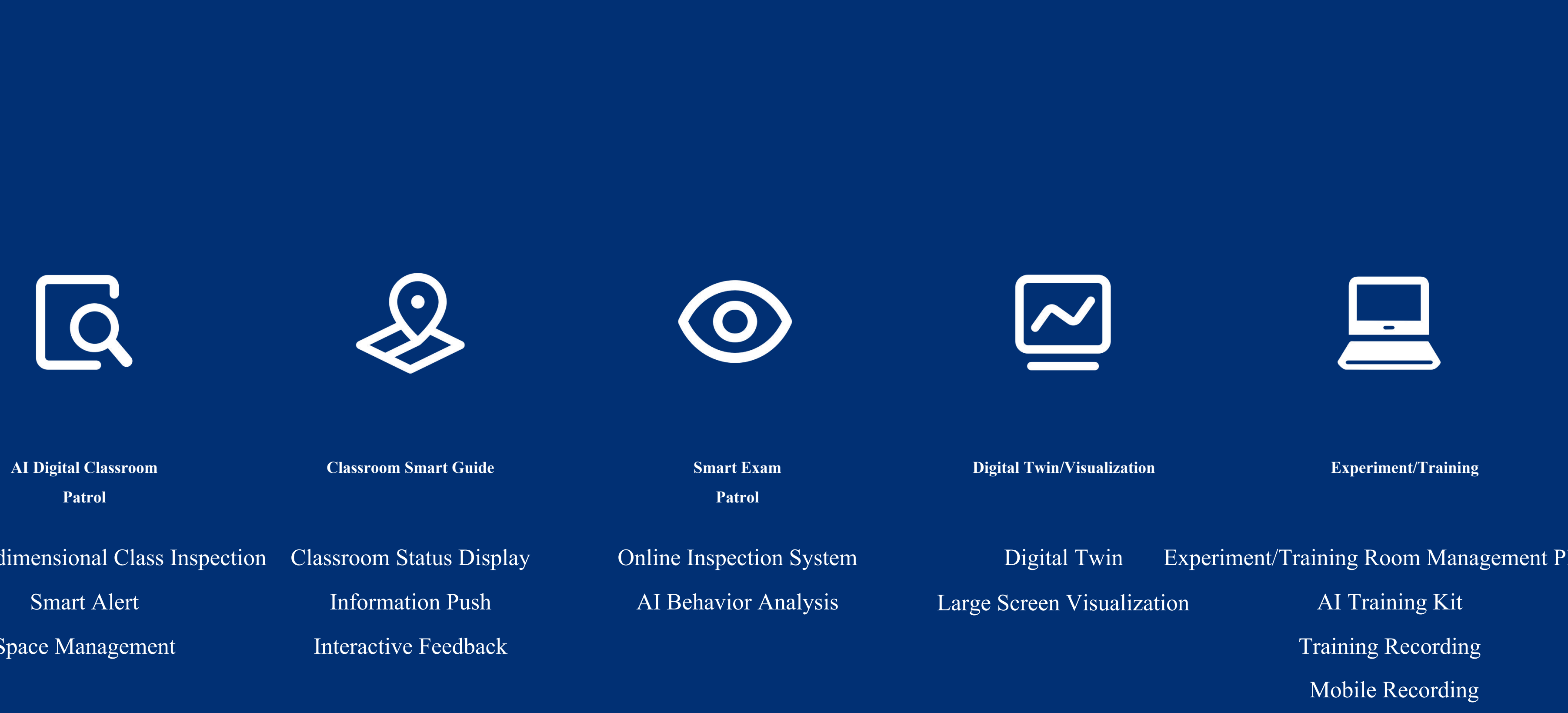


User Profile and Resource Profile

Accurately build user profiles, gain insights into learning behaviors, understand preferences and needs, and provide profile data for personalized learning. Deep analysis of video tags and interaction data, multi-dimensional display of content value, helping to customize learning paths and open a new chapter in smart learning!

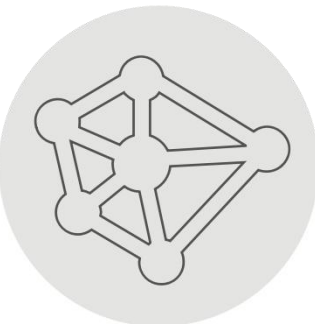
Digital Teaching Support

With the continuous construction of digital campuses, schools are facing challenges such as various types of equipment and difficulties in operation and maintenance management. The digital teaching support system, through technologies and expansion modules such as digital classroom patrols, large-screen visualization, smart classroom navigation, and campus digital twins, can help schools achieve comprehensive management of teaching activities, physical spaces, and information spaces. This enables insights and optimization of teaching resource allocation, reduces the management complexity of teaching support services, assists in improving teaching quality, and builds a multi-level educational digital governance system.



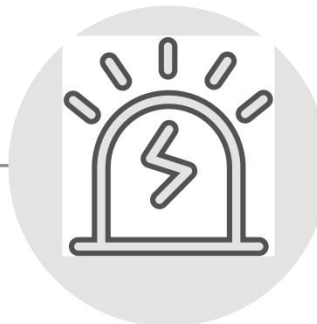
AI Digital Classroom Inspection & Teaching Situation Alert

With the development of educational informatization, traditional teaching management methods can no longer meet the needs of modern educational management and teaching quality improvement. This system aims to utilize media processing and artificial smart technologies to smartly monitor and analyze various teaching spaces such as classrooms, laboratories, and training rooms, providing teaching support departments with more efficient means of maintaining teaching order, and offering teaching quality management departments a more scientific and smart class patrol platform.



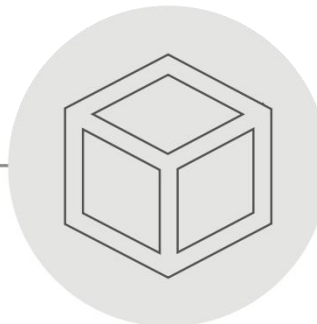
Multi-dimensional Class Inspection

Multi-dimensional class patrol, classroom dynamics at a glance
Screen rotation, teaching scenes under control



Smart Alert

Smart monitoring, rapid warning for abnormal classrooms
Automatic data statistics, effortless operation and maintenance



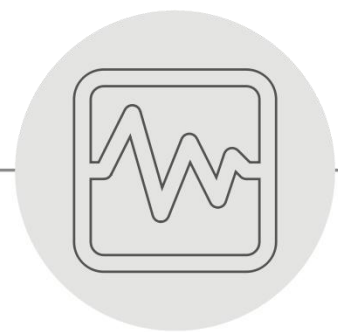
Space Management

Smart statistics, comprehensive grasp of space utilization
Scientific scheduling, optimizing the allocation of teaching facilities



Classroom Smart Guide

The Classroom Smart Navigation System aims to enhance classroom efficiency, optimize the allocation of teaching resources, and provide intuitive classroom status display and interactive services for teachers and students through digital means.



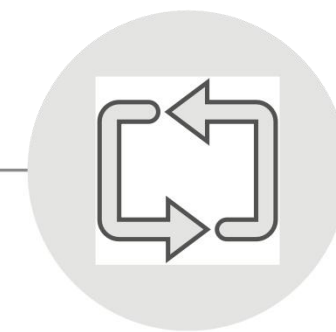
Classroom Status Display

Integrated with the academic schedule, it updates in real time to display the current status of all classrooms in the building, such as in-session, available, study room, or exam, making it convenient for students and teachers to quickly understand classroom usage plans.



Information Push

Integrates information push and campus culture display functions, supporting the release of exam schedules, event previews, and campus news, while showcasing historical, traditional, and special cultural activities, enhancing the dissemination of campus culture.



Interactive Interaction

Supports booking classrooms, querying class schedules and real-time classroom status, automatically detecting time conflicts, providing feedback and tracking processing progress, facilitating the planning and improvement of learning and teaching activities.



Smart Inspection System

Building on the foundation of the National Education Examination Online Inspection System, deploying AI inspection servers can achieve the artificial smart recognition and analysis of abnormal behaviors as required by the [2024]13 letter from the Ministry of Education's Department of Teaching. It enables smart monitoring and real-time analysis of key dimensions such as secure rooms, groups of candidates, individual candidates, and invigilators. By automatically identifying potential abnormal behaviors during the examination process and issuing timely warnings, it ensures the fairness and integrity of the exams.



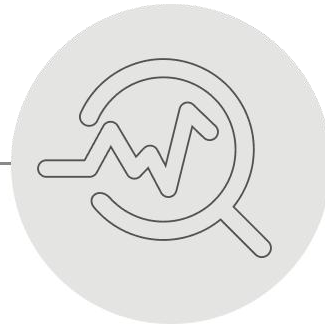
Standard

The system complies with the 'Technical Specifications for Video Standards of the National Education Examination Online Inspection System' (2017 Edition) and GB 36449-2018 'General Requirements for Electronic Examination Room Systems'.



Security

Video storage adopts a distributed + centralized dual backup approach. The equipment complies with GB35114-2017 'Information Security Technical Requirements for Public Safety Video Surveillance Networking.'



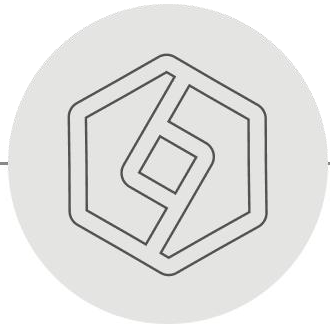
Smart Analysis

Multiple abnormal behavior analysis and early warning are conducted for dimensions such as confidential rooms, groups of candidates, individual candidates, and invigilators, comprehensively ensuring the security of national education examinations and driving examination management towards smart.



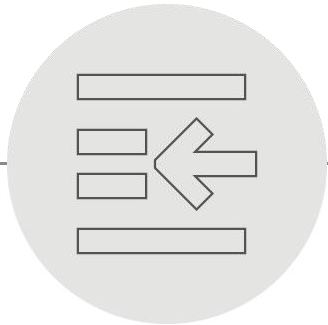
Digital Twin

The campus digital twin system aims to collect data on the school's physical space, attribute information, and geographical location, and then establish its digital twin in the information dimension through technological means. This creates a virtual model of the physical entity in a digital manner, achieving a system that maps and coordinates interaction between the campus's physical space and the network's virtual space. This enables users to achieve comprehensive domain data integration, precision, and smart management of the campus.



Comprehensive Data Integration

Integrate geographic, environmental, and equipment data to achieve precise mapping and real-time interaction of both physical and virtual campuses, providing efficient data support for smart campus management and services.



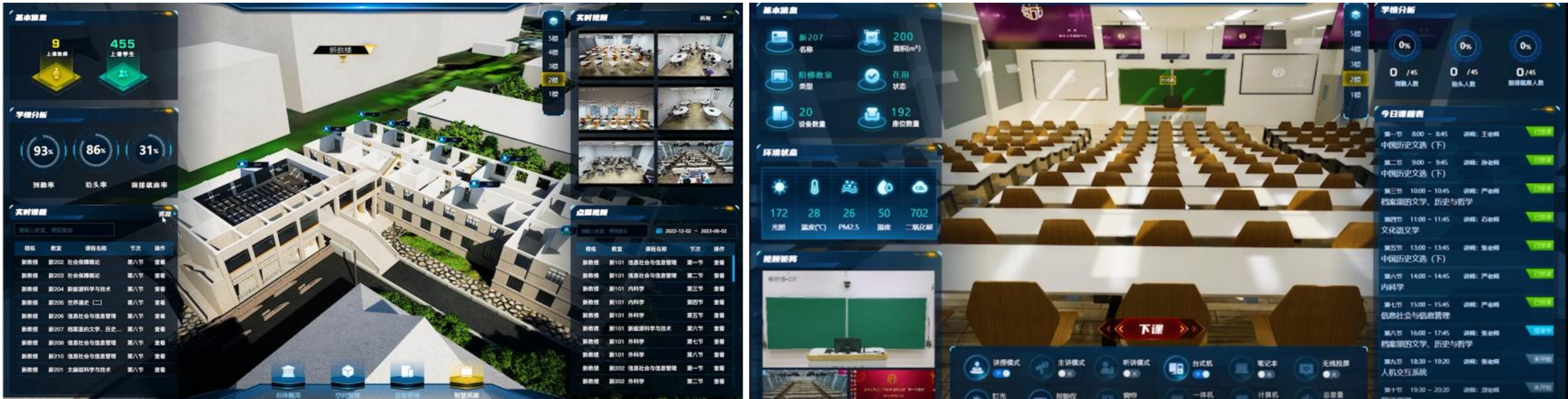
5A System Integration

Can integrate systems such as BA, SA, NA, FA, and TA, achieving predictive maintenance, energy management, and security monitoring through analysis and algorithms, thereby enhancing campus operational efficiency and management levels.



Trend Prediction

By integrating and monitoring data across all domains in real-time, we can accurately capture demands, leverage historical big data, smartly optimize service models, and comprehensively enhance the experience for both teachers and students through trend prediction.



Typical Case Studies

Covering over 30,000 classrooms, including 7 C9 League universities, leading in the number of top-tier domestic university users



Over 40% of Double First-Class Universities

Partial list of Double First-Class Universities (in no particular order)

Tsinghua University

Peking University

Shanghai Jiao Tong University

Zhejiang University

Nanjing University

Fudan University

Xi'an Jiaotong University

Renmin University of China

Tongji University

East China Normal University

East China University of Science
and Technology

Shanghai University

Shanghai Ocean University

Shanghai International Studies

University

Donghua University

Shanghai University of Finance
and Economics

Shanghai University of Sport

Shanghai Conservatory of Music

Shanghai University of Traditional

Chinese Medicine

Chang'an University

Xidian University

Southeast University

Nanjing University of Aeronautics
and Astronautics

Nanjing Agricultural University

Nanjing University of Information
Science and Technology

Nanjing University of Chinese
Medicine

Soochow University

National University of Defense
Technology

Wuhan University

Xiamen University

Shandong University

China University of Petroleum
(East China)

China Academy of Art

Southern University of Science and
Technology...

Double Vocational Education Institutions

Ningbo Polytechnic, Changzhou Vocational Institute of Mechatronic Technology, Wuhu Institute of Technology...

Application Cases

Tsinghua University

Participated in the construction of a smart teaching environment and control platform, providing a highly integrated, fully covered unified video conferencing system. The system includes seven modules: course resources, video conferencing platform, media cloud storage, visual supervision, interactive teaching, electronic examination rooms, and smart analysis. Through a set of classroom video conferencing equipment, it achieves various video business applications such as regular recording and broadcasting, video conferencing, monitoring and inspection, and AI smart analysis. In addition, it integrates existing campus recording and broadcasting systems and classroom security cameras to achieve unified aggregation of classroom resources and centralized platform control.

- 283 New Smart Classrooms
- Regular recording ensures blended online and offline teaching during the pandemic
- Smart multimedia classroom environment management platform
- Integrates with external manufacturers' video conferencing terminals to facilitate international academic exchange
- Artificial smart applications serve teaching management



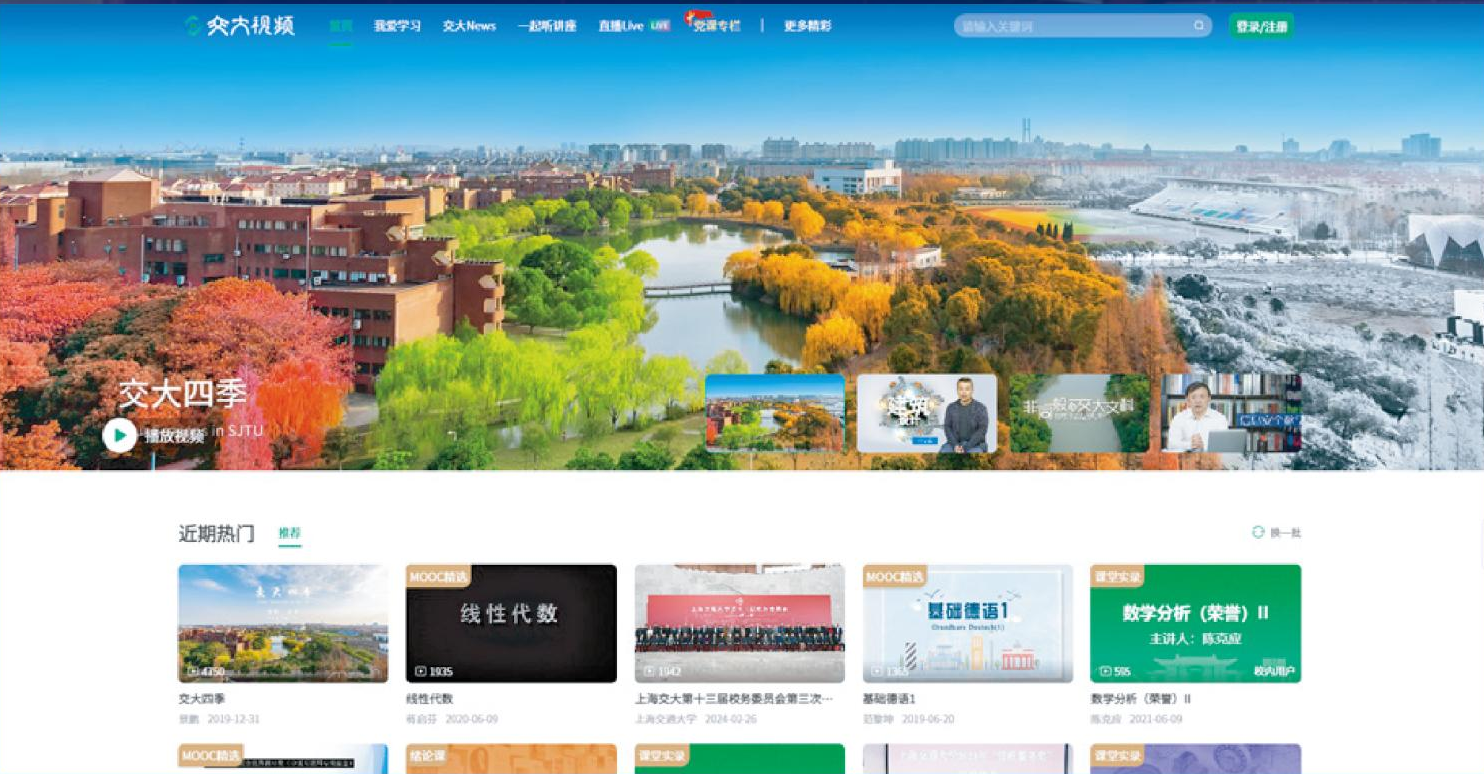


Application Cases

Shanghai Jiao Tong University

Conducted long-term construction planning for Shanghai Jiao Tong University, starting in 2012 with the construction of standardized electronic examination rooms. Through continuous smooth upgrades, the university has achieved full coverage of supervision tours and cloud recording. Post-upgrade classrooms now feature regular recording, multimedia management, teaching supervision, and attendance rate statistics. In 2022, the 'Jiaotong Video' digital teaching resource platform was established, gathering various types of video resources from across the university, now open to the entire network.

- Plan ahead and proceed step by step to protect the school's initial investment.
- Full coverage of 434 cloud-based recording classrooms promotes the construction of school-based resources.
- Full coverage of visual teaching supervision enhances teaching management efficiency.
- AI teaching analysis (attendance rate, personnel redirection) serves teachers and students.
- Support Canvas platform integration and unified portal customization.
- A digital teaching resource platform achieves a unified portal for video resources.



Application Cases

Zhejiang University

Constructed a '1+X' smart classroom video system for Zhejiang University, featuring high system integration and strong reusability. Upon completion, in addition to realizing basic applications such as regular recording, teaching supervision, and remote interaction, it also achieved the expansion of AI applications such as personnel redirection and attendance rate statistics, contributing to the construction of 'Online Zhejiang University'.

- Over 800 4+X type smart classrooms
- Newly built 150 electronic examination rooms
- Exclusive live broadcast support for the 5th China "Internet+" finals
- During the pandemic, 500+ classrooms supported online/offline hybrid teaching
- Explored multi-mode air classrooms such as "ground-to-air" and "ground-to-ground"
- Artificial smart applications empower teaching management



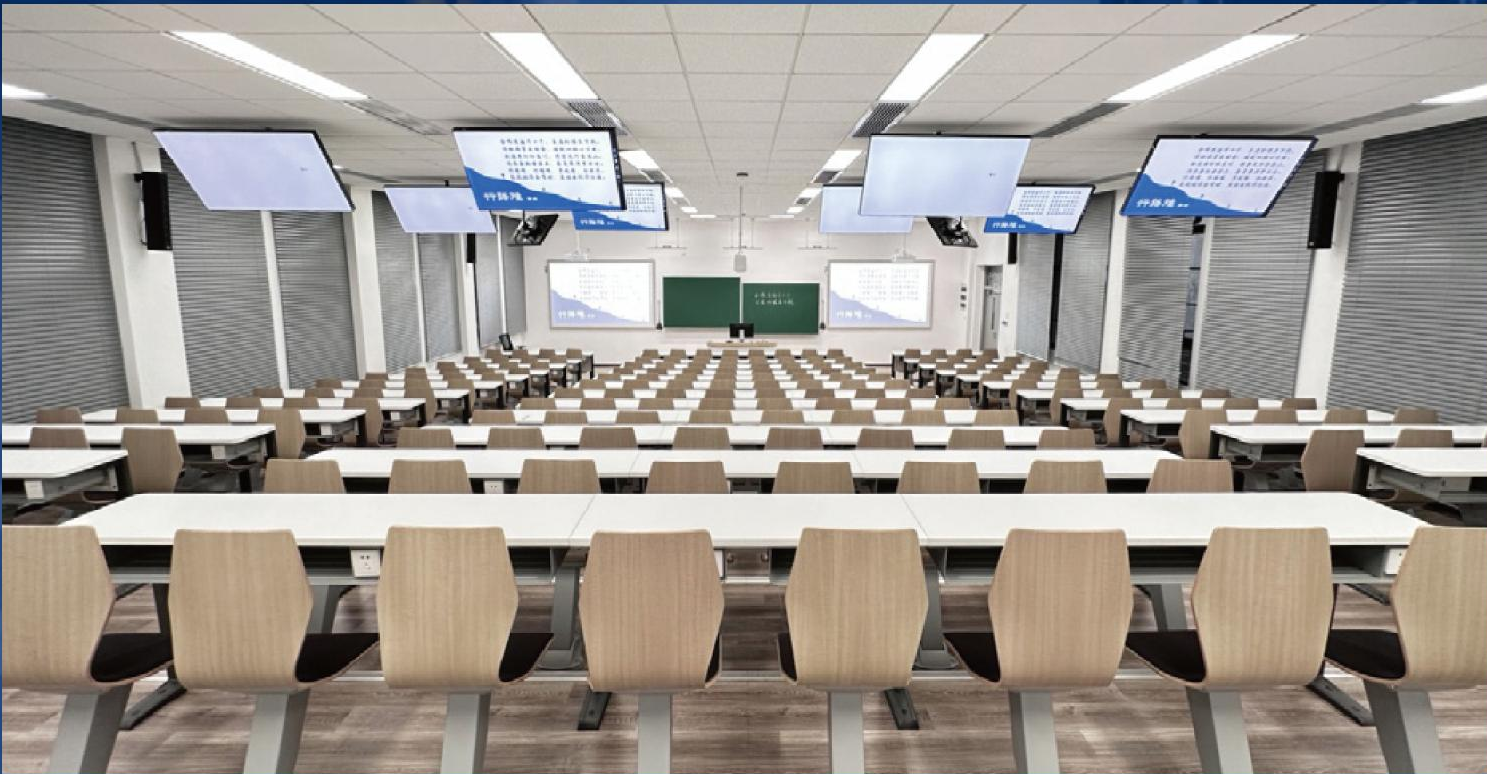


Application Cases

Nanjing University

Based on the "Four-Campus Linkage" theory, multiple smart classrooms were built for Nanjing University, covering Gulou, Pukou, Xianlin, and Suzhou campuses. It provided a technical foundation for the interconnection of teaching resources across campuses, achieving centralized aggregation and unified processing of video resources, ensuring the full integration of online and offline teaching, and helping the university build a campus resource pool.

- Construction of 312 Smart Classrooms
- Remote Interactive Teaching System Across Four Campuses
- Hybrid Online and Offline Teaching
- AI Voice Analysis Engine
- Teaching in Gulou Visualization Center Digital Twin Integrated Management Platform



Selected Representative Cases



Renmin University of China

Covering a total of 160 smart classrooms across the main campus, Tongzhou campus, and Suzhou campus, providing online course resources for students and teachers, and leveraging public cloud CDN for large-scale public live streaming. Building a dual-teacher interactive teaching platform to implement dual-teacher classroom applications across three campuses, and creating a visualization platform using a 2.5D map interaction mode for intuitive and quick switching of large-screen visualizations and data displays. Developing an integrated video communication platform to connect with Jiaclassroom and Rongyou Academy, enabling blended online and offline teaching.



Beijing Technology and Business University

Constructing a total of 79 classrooms in the Liangxiang campus and Fucheng Road campus, assisting the school in building highly integrated and multi-purpose smart classrooms, with the system seamlessly connected to the school's integrated platform to achieve the construction of on-campus resource pools; implementing cross-campus dual-teacher classrooms through interactive recording and broadcasting equipment (including third-party); seamless integration with Tencent Meeting to enable blended online and offline teaching; and integration with the school's security platform to reuse classroom images and surveillance security.



China University of Labor Relations

Construct a total of 102 recording classrooms across the main campus and Zhuozhou campus, deploy an integrated video communication platform to achieve routine resource recording, on-demand course resources, and a supervision system to ensure the evaluation of undergraduate teaching quality. AI-assisted three-rate analysis aids in course evaluation, enabling cross-campus teaching and research activities as well as video conferences based on classrooms. Explore the application of seamless attendance tracking through dual-lens smart cameras for students in classrooms.



China Women's University

Build an integrated video communication platform to integrate and access 60 recording fronts of various brands such as Kedacom, Hanbol, Dongxin Tongbang, and Aoweiya. Connect with third-party on-demand platforms to achieve undergraduate teaching quality assessment and link with the Wanxun operation and maintenance platform for visual operations management. Routine recording classrooms can facilitate blended online and offline teaching, utilizing teaching computers to meet teachers' personalized resource recording needs.



Beijing Institute of Fashion Technology

Starting from 2024, construct cloud recording in 53 public classrooms to achieve high-definition collection of teacher, student, and courseware footage. Deploy an integrated video communication platform and connect with third-party classroom observation platforms to support the university in completing undergraduate education and teaching evaluation tasks.

Selected Representative Cases



Fudan University

In 2020, a teaching supervision system was established, covering a total of 11 classrooms in the Fifth and Sixth Teaching Buildings, integrating 6 existing third-party lecture capture classrooms. The teaching supervision system enables the expert supervision team to conduct online course supervision without time or location constraints. By reusing the equipment and platform of the supervision classrooms, a cloud-based lecture capture system was developed, enabling online live broadcasting and on-demand applications of courses, thereby supporting blended online and offline teaching.



Tongji University

Phase I involves the construction of a video management platform and 600 standardized examination rooms; Phase II expands with 160 additional standardized examination rooms, covering Tongji University's Siping Road, Jiading, Huxi, and Hubei campuses. A two-tier architecture of a master control center and sub-control centers is adopted to establish unified management across the entire area, creating a secure, reliable, real-time, and efficient examination monitoring and command system. In 2021, 173 cloud recording classrooms were newly built or renovated.



Shanghai University

A total of 203 recording classrooms were constructed, repurposed for standardized examination rooms, and deployed with integrated video communication platform, live recording platform, and teaching supervision platform. By localizing the speech-to-text engine, real-time subtitle transcription is achieved during live and on-demand broadcasts. Localized AI services are configured to implement smart functions such as crowd counting, PPT slicing, and OCR recognition.



East China Normal University

The construction includes standardized examination room inspection management system, teaching supervision system, and educational cloud recording system, deploying smart cameras for teachers and students, covering 473 smart classrooms across the school. The system is stable in overall quality, simple to operate, and plays a significant role in teaching video resource construction and management, daily class and examination inspections, and undergraduate teaching review and evaluation.



Shanghai University of Finance and Economics

A total of 114 recording classrooms have been constructed, with 143 third-party recording classrooms connected. A converged video communication platform, recording platform, and teaching supervision system have been deployed to meet the needs of hybrid teaching and class evaluation. The 'SUFU Lecture Hall' video platform has been established to expand the learning boundaries for students.

Selected Representative Cases



Southeast University

Starting from 2012, we have built over 460 video classrooms across the Sipailou, Dingjiaqiao, Jiulonghu, and Wuxi campuses, achieving comprehensive coverage for routine recording, online listening and evaluation of classes, and visual and digital management of classrooms. The early classroom equipment and platform systems have been stably maintained for over 13 years.



Nanjing University of Aeronautics and Astronautics

We established the 'Feitian Cloud Classroom,' constructing more than 280 high-definition recording classrooms. By integrating the video communication center, we connected third-party standard examination and recording equipment, covering 439 classrooms across all three campuses. Video resources are centrally collected and uniformly processed, enabling real-time course broadcasting and post-class video playback, serving over one million teachers and students annually. We also developed a remote interactive teaching system across three campuses, supported by academic performance analysis for digital teaching evaluation and safeguarded the school's review and assessment. The platform integrates DeepSeek, enabling smart classroom summaries.



Nanjing Agricultural University

We constructed an integrated video communication system, building and connecting over 390 recording classrooms across three campuses. At the main Weigang campus, we used hardware online course terminals to enhance the stability of courseware collection for blended online and offline teaching, and customized an online listening platform to improve the listening experience. At the Binjiang campus, we established new smart classrooms, simultaneously building video communication subsystems, teaching quality evaluation systems, and digital patrols, achieving cross-campus connectivity with the main campus system. We also developed a digital twin platform and connected relevant interfaces to enable visual management and control.



Soochow University

We have cumulatively built and connected over 500 recording classrooms, including various teacher training rooms, medical school labs, and 35 smart classrooms, covering multiple campuses such as the main campus, Dushu Lake, Yangcheng Lake, and Future Campus. This enables interactive teaching across campuses, supports remote teaching observation, and facilitates collaborative research across multiple campuses. We have deployed a cloud course resource management platform and a digital lecture observation platform, which are integrated with the university's course resource management system, meeting various business needs such as course recording, live streaming and on-demand viewing, remote interaction, experimental training, and lecture evaluation.



Donghua University

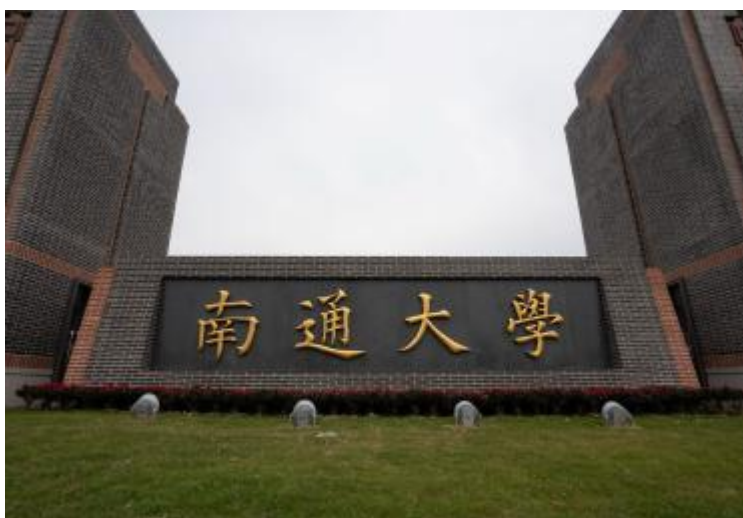
Since 2023, we have cumulatively built 182 cloud recording classrooms, repurposed for standardized exam scenarios, effectively improving equipment utilization. We have deployed a teaching supervision cloud platform, repurposed for various applications such as class observation, supervision, and leadership attendance, energizing digital video resources in multiple areas including teaching, management, evaluation, and examination.

Selected Representative Cases



Jiangsu University

Develop an online and offline hybrid teaching system, deploy high-performance distributed media storage and high-density servers, establish 60 new classrooms and connect 124 existing classrooms with front-end video equipment, providing various applications such as course live streaming and on-demand viewing, media asset management, supervision management, visual management, and comprehensive data analysis. Fully utilize existing resources to precisely empower digital transformation.



Nantong University

Starting in 2023, the video platform will be upgraded, with 180 Kedao recording classrooms being built in batches, integrating and connecting over 500 classrooms of various types across three campuses. The 'Nantong University Cloud Classroom' video comprehensive platform will be deployed, including applications such as classroom recording, Nantong University videos, online course auditing, AI learning analysis, AI voice, and digital course patrol, to enhance teaching management and quality improvement.



Zhejiang University City College

Constructed 200+ smart classrooms to serve online/offline hybrid teaching, pushing content to course platforms for live streaming while also locally recording backups to prevent data loss; provided a teaching supervision platform to enhance the school's information technology level and supervision efficiency; offered smart analysis services such as attendance statistics, personnel flow guidance, seating rate, and engagement rate, solving the issues of class roll call and student attendance management for teachers.



Zhejiang Chinese Medical University

Remodeled a total of 170 public classrooms in the Binwen and Fuchun campuses by installing high-definition recording hosts, courseware collectors, and teacher cameras in the classrooms to collect basic audio and video data of teachers and courseware during the teaching process. This enabled routine recording and video archiving, online live streaming, and on-demand playback, seamlessly integrating with software such as DingTalk and Tencent Meeting to achieve online/offline hybrid teaching.



Zhejiang University of Finance and Economics

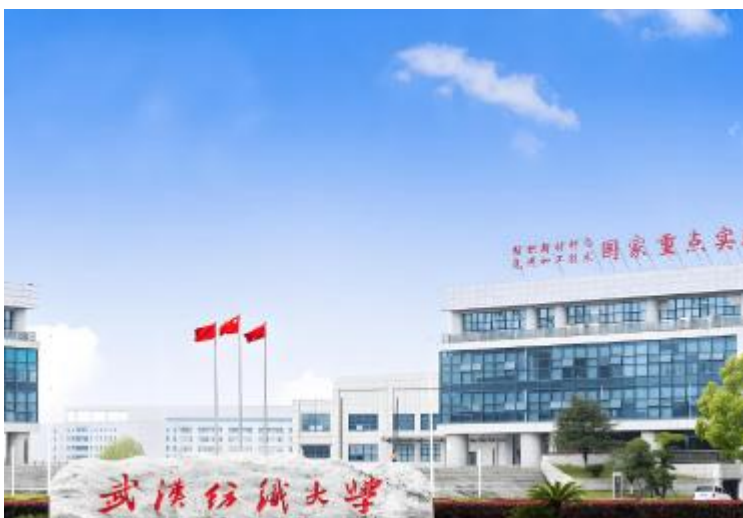
Construct 156 cloud recording classrooms, equipped with high-definition video encoders and teacher cameras. Combined with third-party online meeting software, it achieves hybrid online and offline teaching. Additionally, it integrates with the Chaoxing resource platform to enable one-click uploading of course resources and data interoperability.

Selected Representative Cases



Wuhan University

Establish a new integrated video platform and teaching supervision system, achieving full coverage of over 640 classrooms across the campus. This includes constructing 28 new recording classrooms and integrating over 620 existing classrooms with third-party brands. Through the supervision listening and evaluation system, along with on-site support services provided by the original manufacturer, the university successfully conducted online listening and evaluation tasks for undergraduate teaching assessments using the full opening of over 640 classrooms.



Wuhan Textile University

Construct 172 cloud recording classrooms, 10 smart classrooms, 1 interactive smart classroom, and 5 video conference rooms for holding meetings and conducting cross-campus courses at Nanhu Campus and Yangguang Campus. Build an integrated video platform covering services such as routine recording, online supervision, real-time class monitoring, learning analysis, and conference interaction, to assist in the smooth completion of undergraduate teaching evaluation.



Shandong University

Establish a smart classroom video system in 51 classrooms at the Weihai Campus, enabling the routine recording of course resources, online supervision and evaluation of teaching, and automatic attendance statistics through AI image analysis technology. Additionally, 2 centralized control rooms were built to achieve visual centralized management of classrooms, and the equipment is reused for daily security monitoring in classrooms.



China University of Petroleum (East China)

Transform 269 online interactive classrooms, providing in-depth support and integration for existing multi-brand, multi-type audio-visual resources and equipment. Establish an online interactive teaching system for the Tangdao Bay and Ancient Town campuses, creating a visual teaching management system to achieve large-scale classroom inspections, convenient class supervision, and immersive class attendance. Build the ShiDa Lecture Network, unifying the Huangdao Forum and high-quality courses.



Anhui Normal University

Develop a comprehensive teaching resource integration platform, repurposing and connecting 92 classrooms across the Huajin, Zheshan, and Tianmenshan campuses, enabling multi-campus interaction to meet high-concurrency demands for live and on-demand broadcasting. Additionally, provide standard interfaces to support future new business systems for the school. Deploy a classroom smart navigation system, facilitating easy access to schedules and classroom locations for teachers and students. Install new large model integrated machines, integrating multi-modal large models like DeepSeek, to achieve comprehensive classroom element collection, optimize class attendance experience, and utilize multi-dimensional data analysis to enhance teaching management and decision-making.

Selected Representative Cases



Xiamen University

Develop a '1+X' smart classroom video application system with high integration and strong reusability. Upon completion, in addition to realizing basic applications such as routine recording, online live streaming, and on-demand viewing, it also expands various AI applications such as AI teaching supervision and attendance rate statistics, contributing to the construction of the 'Digital Teaching Platform'.



Jinan University

Build a teaching video management platform and teaching inspection system, connecting upwards to the Guangdong Provincial Education Examination Institute platform, covering 317 classrooms across the entire campus, and achieving standardized electronic examination rooms and daily class inspections, teaching observations, and other applications.



Southern University of Science and Technology

Build a school-level recording and broadcasting management platform, establish 72 smart classrooms, equip the school with high-performance storage and high-density servers, provide real-time course livestreaming, on-demand playback after class, leadership attendance, resource management and other application services, equip with an integrated large model machine, and integrate AI capabilities such as DeepSeek to comprehensively enhance the collection, management, and digital application of course resources.



Chang'an University

Batch transform and build 300 regular recording and broadcasting classrooms, and reuse them as standardized examination rooms. Integrate with the Zhiyuan learning platform, actively cooperate with the school during the pandemic to achieve 'no suspension of learning despite class suspension' through online teaching, continuously plan and iterate new business capabilities, and empower the school's digital transformation.



Xidian University

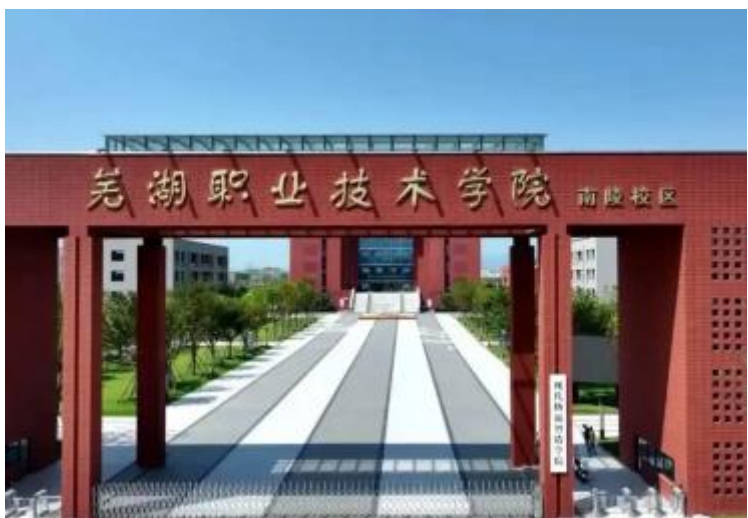
Build over 400 standardized electronic examination rooms, deploy a video inspection system, and integrate it with the superior examination institute's platform. Equip more than 17 high-performance cloud storage devices to preserve and manage examination room videos and educational materials. Additionally, build an academic seminar and training space information system based on a video conferencing solution, enabling remote video conferences across four campuses, and establishing a remote communication platform for international universities and scholars.

Selected Representative Cases



Ningbo Polytechnic

Develop a digital course patrol system to enable real-time classroom monitoring. Additionally, utilize smart analysis servers to collect data such as front-row seating rate, attendance rate, and attention rate during class hours. Reuse the dual-lens smart cameras from the patrol system to manage student attendance in classrooms during class hours.



Wuhu Institute of Technology

Phase-by-phase construction of over 100 video classrooms across the South and North campuses, creating a high-quality teaching resource library for the university, enabling future extensions such as teaching supervision, and enriching teaching management methods. Utilize and integrate third-party high-quality recording classrooms, enabling interactive teaching applications between classrooms through recording hosts, achieving a new cross-regional teaching model.



Changzhou College of Mechanical and Electrical Technology

Build 66 new recording classrooms, covering the Science and Education City Campus and Liyang Campus; seamlessly integrate 28 existing surveillance cameras to centralize video resources; establish a new online teaching observation platform, combined with AI learning analysis to visualize the teaching process, effectively solving the inefficiency and lack of data statistics in traditional supervision; configure mobile recording kits to support the recording of videos for teaching competence competitions in vocational colleges.



Suzhou Vocational University

Implement the 'Sunshine Sports' solution to meet the needs of students for running exercises and card swiping; install LED small-pitch large screens in the school's conference rooms, training rooms, and smart campus command centers to enhance the audio-visual experience in multiple scenarios such as meetings, training, command, and centralized class observation.



Nanjing City Vocational College

More than 100 multimedia classrooms, banquet halls, and conference rooms have been renovated at the Licheng and Xinjiekou campuses, with the deployment of an integrated video platform to enable regular course recording. Additionally, based on the school's needs, more than ten production training rooms have been constructed, creatively empowering modern and personalized teaching scenarios.

Service and Support

Seacraft adheres to the service philosophy of 'customer-first, creating value for customers,' and has established a professional localized support team dedicated to providing clients with fast-response, flexible, and efficient service experiences. Leveraging solid technical expertise, innovative service thinking, and a 'on-demand' support strategy, the team offers customized and comprehensive solutions to help clients achieve value enhancement. Seacraft aims to become a trusted long-term partner for clients through precise service delivery and continuous technological upgrades.



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